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| **Project Overview** | |
| Objective | Develop an intelligent machine learning system that predicts whether a credit card application should be approved or rejected based on applicant information with high accuracy. |
| Scope | The project includes data collection, preprocessing, feature engineering, model training, evaluation, and deployment using Flask. Multiple machine learning algorithms are compared to identify the best-performing model for real-time prediction. |
| **Problem Statement** | |
| Description | Banks receive thousands of credit card applications every day. Manual verification is time-consuming, inconsistent, and prone to human error, resulting in delayed approvals and increased operational costs. |
| Impact | Automating the approval process improves decision-making speed, increases prediction accuracy, reduces financial risks, minimizes manual effort, and enhances customer satisfaction. |
| **Proposed Solution** | |
| Approach | Historical applicant data is pre-processed and used to train Machine Learning models including Logistic Regression, Decision Tree, Random Forest, and XGBoost. The best-performing model is integrated into a Flask web application for real-time prediction. |

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| Key Features | • Automated Credit Card Approval Prediction<br>• High Accuracy using Machine Learning<br>• Real-Time Prediction through Web Application<br>• User-Friendly Interface<br>• Fast Decision-Making Process<br>• IBM Watson Machine Learning Deployment Support<br>• Secure and Scalable Architecture |

## Project Initialization and Planning Phase

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| **Date** | June 28, 2026 |
| **Team ID** | SWTID-2026-8797 |
| **Project Title** | Credit Card Approval Prediction using Machine Learning |
| **Maximum Marks** | 5 Marks |

**Project Proposal (Proposed Solution) Report**

This project automates the credit card approval process using machine learning. It analyses applicant financial and demographic information to predict whether a credit card application should be approved or rejected. The solution reduces manual effort, improves consistency, speeds up decision-making, and minimizes the risk of human error.

**Resource Requirements**

| Resource Type | Description | Specification/ Allocation |
|---|---|---|
| **Hardware** | | |
| Computing Resources | CPU/GPU specifications, number of cores | Intel Core i5/i7 Processor (4+ Cores) |
| Memory | RAM specifications | 8 GB RAM (Minimum), 16 GB Recommended |
| Storage | Disk space for datasets, trained models, and application files | 512 GB SSD / 1 TB HDD |
| **Software** | | |
| Frameworks | Python Web Framework | Flask |
| Libraries | Machine Learning and Data Processing Libraries | Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost |
| Development Environment | IDE | Jupyter Notebook, Visual Studio Code, PyCharm |
| **Data** | | |
| Data | Source, Size, Format | Credit Card Approval Dataset (Kaggle), CSV Format, ~20,000+ Applicant Records |